

Designing the Future of Packaging:
Building Circular and Sustainable Systems for Everyday Products

Trace & Reward: Waste Traceability via SGTIN and Integrated Shopee Ecosystem Gamification

“Order Arrives, Waste Bye-bye!”



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Executive Summary

Indonesia produces more than **37 million tons of waste** per year with plastic packaging ranking as the **second-largest contributor** to this environmental crisis. In response, Nestlé Indonesia, one of the biggest manufacturing companies, aims to achieve **Net Zero emissions by 2050** and transition to sustainable packaging. While Nestlé has successfully **replaced plastic straws with paper** (excellent package design) and established **Waste Stations**, a critical circular packaging gap remains: **Traceability Issue**. Ideally, the product journey should be circular. However, once a product **leaves the Point of Sale (POS)**, Nestlé loses **traceability**, resulting in **low collection rates**. This is exacerbated by Indonesia's waste landscape where **only 32% of waste** across 342 cities is formally processed.

To bridge this **traceability problem**, we introduced an approach called **Trace & Reward**: a local waste traceability ecosystem integrated with **Shopee** because **most Indonesians**, ranging from low-income to high-income, conduct a transaction online via their services. By transforming the standard SGTIN into a **Serialized Waste Identity**, we enable a “**Scan-to-Earn**” mechanism. Consumers can now seamlessly return packaging via **Reverse Logistics** (piggybacking on e-commerce deliveries) or verified Drop Points, earning **Shopee Coins** or rewards as an instant incentive.

This approach aligns with and combines “**developing new reuse, refill, or collection systems**” with “**leveraging technology or data to improve packaging traceability**” scope and creates a **closed-loop** system where **waste becomes currency**. It empowers Nestlé and other manufacturing companies with real-time **post-consumer data**, drastically boosts **collection rates** for high-quality recyclables and **deeply engages Indonesia's digital-native youth** with rewards. Therefore, Trace & Reward turns the passive act of disposal into an **active contribution** to Nestlé's **Net Zero** roadmap by turning **pain points** (waste management friction) into points, **reducing untraced waste** by up to **53% at best scenario** or **44% at conversion rate** passing the 30% requirement in Permen LHK No. 75 Tahun 2019.

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Chapter I: Introduction

1.1 Background and Current Situation

Indonesia is witnessing a significant shift in consumer behavior driven by the "Green Generation", primarily Gen Z and Millennials who prioritize sustainability in their purchasing decisions (IDN Research Institute 109). To address this, Nestlé has committed to ambitious waste reduction targets through its "needs YOUth" and global sustainability initiatives. However, a critical challenge remains in the post-consumer journey: packaging waste becomes untraceable after the Point of Sale. While Shopee currently dominates the Indonesian e-commerce landscape with a 53% primary transaction share (GoodStats), actual recycling rates remain low due to system friction. Consumers often find existing recycling infrastructure requires "too much" effort (GoodStats), creating a gap between high environmental awareness and low actionable participation.

1.2 Complication and Problem

This lack of traceability is further complicated by increasing regulatory pressure. Manufacturing companies like Nestlé must comply with Extended Producer Responsibility (EPR) regulations to meet government waste reduction targets, avoid legal penalties (Peraturan Menteri LHK No. 75 Tahun 2019), and support sustainability. Currently, there is no efficient mechanism to track *unique* product packaging back to the manufacturer after consumption. This "Traceability Gap" prevents accurate impact measurement and makes it difficult to reward users for responsible disposal. Consequently, recyclable materials frequently end up in landfills rather than being reintegrated into the production cycle, hindering the transition to a circular economy (Sustainability Directory).

1.3 Solution and Future State

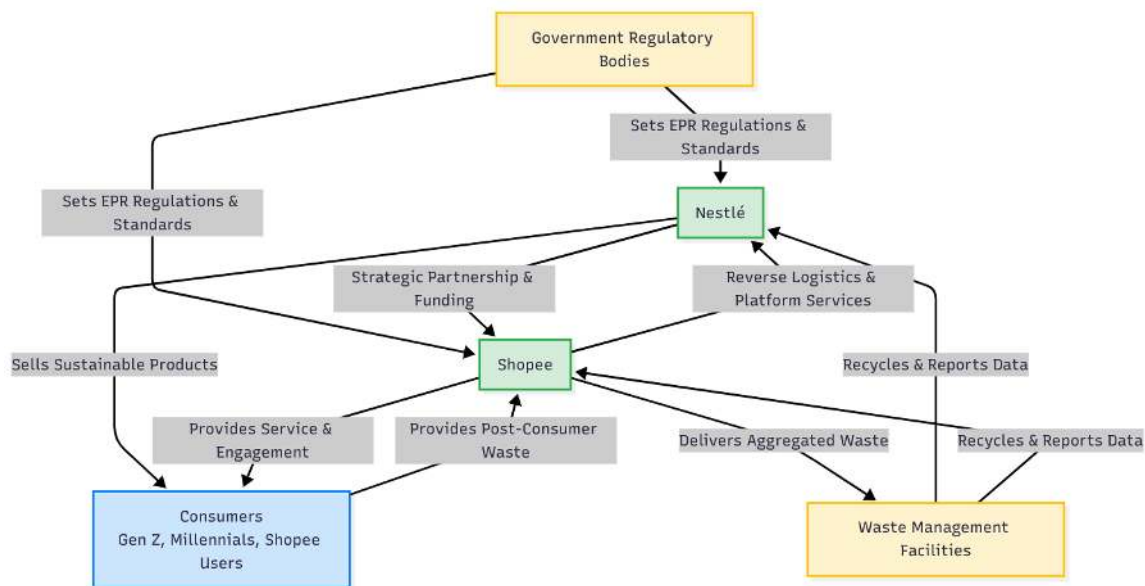
This proposed solution by Team Lunaiki resolves these friction points by transforming every piece of Nestlé packaging into a *digital asset*. By implementing SGTIN instead of GTIN and integrating QR-based traceability directly into the massive Shopee ecosystem, we can bridge the distance between individual consumers and recycling facilities. In this envisioned future, consumers can "trade" waste via the Shopee app to receive tangible rewards, such as Shopee Coins or vouchers. This gamified approach resolves the lack of consumer motivation, while the backend data provides Nestlé (as the "initiator" for other manufacturing companies) with real-time transparency for EPR compliance. Ultimately, this creates a closed-loop system that mitigates legal risks and strengthens Nestlé's position as the leader in the sustainable FMCG market.

Chapter II: Stakeholder & Problem Analysis

2.1 Stakeholder Relations

2.1.1 High Impact/High Influence Stakeholders

Here is a simple diagram showing the relations between stakeholders where Nestlé is the focus company in this proposal.



2.1.2 Mapping Collaboration between Stakeholders

A business model canvas to capture the relations can be shown below.

Key Partners	Key Activities	Value Proposition	Customer Relationships	Customer Segments
Shopee Provides the digital platform (App), gamification engine, and reverse logistics (Shopee Express). Nestlé Provides the products, packaging with SGTIN codes, and sustainability funding. Waste Management Facilities Partners who actually recycle the material collected by Shopee Express.	1. Coordinating Shopee Express to pick up waste when delivering orders. 2. Designing leaderboard & rewards for collections. 3. Ensuring collected waste is clean and recyclable. Key Resources 1. The App (User Interface) and Shopee Express. 2. Adapt SGTIN Technology 3. Nestlé Brand Power: Trust and market penetration. 4. User Insights on consumption patterns via User Data.	Consumers 1. "Order Arrives, Waste Bye-bye." No need to travel to recycle; do it from your doorstep. 2. Earn Shopee Coins/Discounts for every item recycled. 3. Real-time tracking of waste impact via the app. For Nestlé/Shopee 1. EPR Compliance: Meets government waste reduction targets. 2. Brand Loyalty: Positions both brands as sustainability leaders.	1. Users scan and schedule pickups entirely within the Shopee App. 2. Gamification creates a sense of competition (leaderboards) and shared goals among users. 3. Recurring rewards system motivates users. Channels 1. Shopee Mobile App 2. Shopee Express Couriers as channel for waste collection (Rev. Logistics) 3. Social Media for promoting "Green Generation" campaign.	Gen Z & Millennials as the primary drivers of the "Green Generation." Existing Shopee Users (53% of Indonesians) who use Shopee as their main transaction platform. Nestlé Product Consumers and families who buy FMCG products regularly.
Cost Structure		Revenue Streams		
1. Logistics costs for fuel and labor for Shopee Express to handle reverse pickups (minimized by combining with deliveries). 2. Platform Development: API integration between Nestlé SGTIN database and Shopee App. 3. Cost of Shopee Coins or discounts given to users. 4. Marketing and campaign costs to educate users on how to recycle		1. Increased sales volume as users prefer Nestlé products to earn "Recycling Points" on Shopee. 2. Material Recovery Value: Selling sorted, high-quality recyclable materials to processors. 3. Data Monetization in valuable insights on post-consumer behavior (who recycles what and when). 4. Reduced EPR penalties to avoid potential government fines for waste non-compliance.		

2.2 Root Cause Analysis

Why do the problems occur?



The problem emerges with an *information black hole*. Currently, Nestlé and other manufacturing products effectively vanish from the company's data ecosystem immediately after the Point of Sale (POS). This lack of post-consumer visibility makes it impossible for them to track the *lifecycle of its packaging* or verify the 30% waste reduction mandated by and based on Permen LHK No. 75/2019 to support sustainability.

This invisibility persists because there is no digital bridge, such as a unique SGTIN or QR-based tracking system, to connect physical packaging back to the manufacturer after it reaches the consumer. Without a tracking mechanism, the packaging remains "dumb" waste rather than a *digital asset* that can be accounted for in a circular economy.

Because no digital bridge exists, the burden of recycling falls entirely on the consumer. For the target Gen Z and Millennial demographic, the current manual process of finding waste banks or drop-off points is perceived as requiring "too much" effort. *High environmental awareness* is currently neutralized by *high physical friction*.

The friction is further compounded by a lack of tangible motivation. There is currently no integrated reward hub within the daily digital platforms already used by 53% of Indonesians. By failing to meet consumers where they already spend their time—specifically on Shopee—the system misses the opportunity to transform a "chore" into a rewarding habit.

The current system is designed for a one-way trip: from production to landfill. It lacks the platform-driven feedback loops necessary to *close the circle*.

2.3 System Requirements Analysis

2.3.1 Stakeholder Requirements

For the solution to be implemented properly, we need to specify what requirements each stakeholder gets addressed from the system.

Stakeholder	Requirements
Nestlé (Manufacturing Company)	1. Access to a real-time dashboard displaying verified recycling volumes and SGTIN-level data.
Shopee (Rev. Logistics and “Motivator”)	1. The system must integrate seamlessly with existing Shopee Coins and Voucher APIs for automated reward distribution. 2. Penalty system for falsifying waste and incentives for drivers.
Driver (Logistics)	1. Minimum weight and maximum weight to ensure feasibility. 2. Must differentiate between delivery to the user and waste to the waste management so that the delivery is not dirtied. 3. Incentive/reward for collaborating in the program.
User (Consumer)	1. Scan packaging in under 3 seconds with immediate visual confirmation of promise for rewards. 2. Receiving orders and sending the waste to the driver in under 30 seconds at doorstep.

2.3.2 System (Closed-Loop) Transition and Quality

1. Security & Data Integrity

The system must utilize unique SGTIN/QR-based traceability to prevent "double-scanning" or fraudulent reward claims. All recycling data must be encrypted to ensure its validity for government and user rewards.

2. Scalability

The backend infrastructure must support simultaneous scans from millions of active users during system implementation.

3. Interoperability

The data bridge must be compatible with Nestlé’s existing supply chain ERP and Shopee’s logistical ecosystem.

4. Migration

“Update” the current implementation from GTIN to SGTIN and data collaboration between Nestlé and Shopee.

5. Information

An education about how to use the scan system in Shopee.

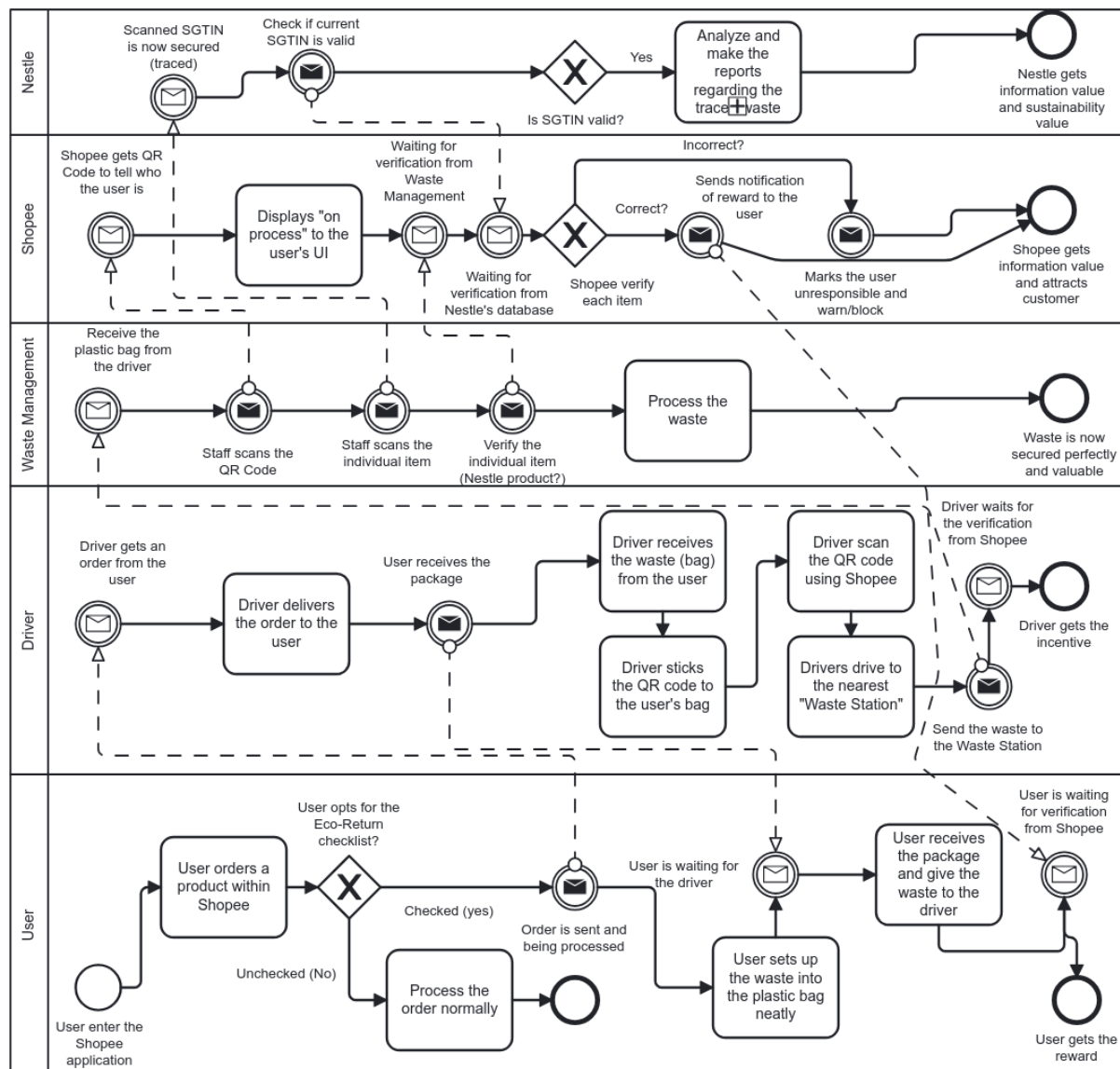
Chapter III: The Proposed Solution

3.1 The Proposed Solution Summary

To overcome the "too much effort" barrier identified in the 2.2 Root Cause Analysis, this solution integrates waste traceability into the daily shopping habits of the Green Generation through shopee. The solution transforms the chore of recycling into a seamless, rewarding digital experience by using Shopee and its ability to support reverse logistics through Shopee Express.

3.2 Diagram Model of The Proposed Solution

Here is a simplified Business Process Model and Notation for the mechanism.



The process starts when a user enters the Shopee application and places an order. They are presented with an exclusive gateway to opt into the "Eco-Return" checklist after they choose to order. If the user declines, the order is processed normally. However, if they opt-in, they can begin preparing plastic waste in a bag while the order is being processed and dispatched.

Next, the logistics phase begins when the driver receives the order and delivers the package to the user's location. During this exchange, the user receives their new package and simultaneously hands over the prepared waste bag to the driver. The driver then performs a tracking step by sticking a unique QR code onto the user's bag and scanning it via the Shopee app. Following the collection, the driver transports the waste to the nearest "Waste Station" and awaits verification from the platform to receive their incentive for successful delivery .

The verification and processing phase occurs at the Waste Management facility where staff receive the bag and scan both the bag's QR code (manually or using an automaton) and the individual waste items. This data is cross-referenced to verify if the items are Nestlé products, sending a signal to Nestlé to check if the Serialized Global Trade Item Number (SGTIN) is valid: SGTIN exists and not yet collected. If the SGTIN is valid, Nestlé analyzes the data to generate sustainability reports, thereby gaining significant information and sustainability value. Simultaneously, Shopee waits for verification from both the Waste Management staff and Nestlé's database to ensure the returned items are indeed correct and valid.

Lastly, Shopee evaluates the verification results through another decision gateway. If the items are found to be incorrect or fraudulent, the user is marked as irresponsible and may be warned or blocked from the program. If the verification is successful, Shopee sends a reward notification to the user and an incentive to the driver. The process concludes with a Win-Win outcome as the user receives rewards, the driver earns an incentive, the waste is secured as a valuable resource, Shopee and manufacturing companies enhance their ESG (Environmental, Social, and Governance) value to attract customers, and companies achieve measurable traceability for its packaging lifecycle. This removes the "Traceability Gap" problem.

Chapter IV: Strategy and Risk Analysis

4.1 SWOT & TOWS Analysis

Before the risk analysis is mentioned, this is the proposed solution's strengths, weaknesses, opportunities, and threats table to show what is great and what is not.

Strengths (Internal)	Weaknesses (Internal)
1. Shopee's 53% share provides instant user access. 2. Uses the existing Shopee Express fleet for reverse logistics (low cost). 3. Unique codes ensure accurate tracking and fraud prevention.	1. Risk of users returning dirty/unrecyclable items. 2. High initial cost to sync Nestlé's SGTIN DB with Shopee App. 3. Additional workload for drivers may cause operational friction.
Opportunities (External)	Threats (External)
1. The majority of Gen Z/Millennials actively seek sustainable options. 2. Selling materials to waste facilities as a revenue stream. 3. Users prefer Nestlé products to earn Shopee rewards.	1. Government fines for failing waste reduction targets. 2. Drop-off in participation if the process is too complex. 3. Cyber security threats in cross-company data sharing.

To answer the challenges in the weaknesses and threats section, the TOWS matrix is provided below.

TOWS	Threats	Opportunity
Weaknesses	1. Restrict pickups to "Delivery Only" (piggyback model) to control logistics costs. 2. Launch a pilot region first to fix API bugs before national scale.	1. Use app games to teach users how to clean waste, improving material quality. 2. Share a percentage of material revenue with drivers to motivate waste collection.
Strengths	1. Use SGTIN data to generate precise reports for government audits, avoiding EPR fines. 2. Use app verification to ensure only valid Nestlé packaging counts toward targets.	1. Leverage Shopee's user base for "Trade & Win" challenges to capture the Gen Z market. 2. Use SGTIN tracking to identify and sell high-value recyclables.

4.2 Implementation of The Current System Strategy

4.2.1 Quality Fail-safes

To effectively manage the identified risk of users returning dirty or unrecyclable items, the system implements a "Visual Inspection" protocol during the physical exchange between the user and the driver. Drivers are empowered to refuse waste bags that show visible leakage or contamination to protect the primary delivery mission of the Shopee Express fleet. This fail-safe ensures that only high-quality materials and packing (trash bag) reach the Waste Management facility for processing.



4.2.2 Data Governance and Privacy Framework

Addressing the high initial cost and technical complexity of syncing Nestlé's or other manufacturing companies' SGTIN database with the Shopee App requires a secure API integration. To mitigate cybersecurity threats during this cross-company data sharing, all recycling data must be encrypted to ensure its validity for both government audits and user reward distribution. The system relies on a unique SGTIN/QR-based traceability to prevent "double-scanning" or fraudulent reward claims. By maintaining a digital bridge that only shares necessary verification signals, the framework protects sensitive consumer data while providing Nestlé with the real-time transparency needed for EPR compliance and sustainability reporting.

4.2.3 Pilot-to-Scale Transition Strategy

The implementation roadmap follows a phased approach, launching in a specific pilot region first to identify and fix API bugs before expanding to a national scale. This initial phase focuses on high-density urban areas where the Shopee Express fleet and Waste Management partnerships are already established. During the pilot, the "piggyback model" will be strictly enforced. This restricts waste pickups to "Delivery Only" windows to control logistics and fuel costs. Success in the pilot stage will be measured by the system's ability to minimize behavioral friction for the Green Generation. Once the model is stabilized, it will scale nationally to help Nestlé and other manufacturing companies meet the 30% waste reduction targets mandated by government regulations, leading to more eco-friendly products.

4.3 Addressing Secondary and External Risks

4.3.1 Conflict with the Informal Sector

Indonesia has a massive informal waste-picking economy (*pemulung*) that relies on collecting high-value plastics like PET from households (Frigo et al.). By diverting this waste directly into a digital "Shopee-Nestlé" loop, the program risks displacing the livelihoods of vulnerable communities.

Therefore, transition from a "competitor" to a "partner" by allowing local waste banks or registered collectors to act as verified aggregators is needed. These aggregators can use the Shopee App to bulk-scan collected SGTINs and receive a service fee that standardizes, formalizes, and increases their income.

4.3.2 The "Rebound/Cobra-like Effect"

There is a risk that the gamification of waste—turning packaging into "currency"—may inadvertently encourage consumers to purchase more single-use products specifically to earn Shopee Coins. Although this would increase Nestlé's products' sales, this rebound effect would increase Nestlé's total plastic footprint, contradicting the Net Zero 2050 goal.

To mitigate this, we can implement Capped Monthly Rewards. Once a user reaches a certain sustainability tier, rewards shift from per-item coins to "Eco-Education" badges or discounts on Nestlé's refillable or bulk-buy products, nudging consumers toward truly zero-waste habits.

4.3.3 Connectivity Risk

While Shopee has high penetration, internet stability varies across Indonesia's 17,000 islands (Braham). A "Scan-to-Earn" system that requires constant real-time synchronization may fail in rural or low-connectivity areas. This can cause user frustration and loss of trust, leading to the thickening of the "barrier to continue".

The application will support offline data logging. QR codes can be scanned and stored locally on the user's or driver's device. The "Pending" points will automatically sync and verify with the SGTIN database once the device has a stable connection, for example entering a stable 4G/5G zone.

4.3.4 Privacy and Data Trust Risks

Consumers may feel *anxiety* about sharing consumption data with two large corporations. This is a privacy issue that needs to be addressed.

Therefore, the Shopee-Nestlé data bridge will utilize anonymized aggregation where Nestlé receives high-level geographical waste trends, for instance 500 kg collected in Bandung, rather than individual user-linked consumption profiles.

Chapter V: Financial and Feasibility Analysis

5.1 Cost-Benefit Analysis

Financial Component	Unit Value (in IDR)	Total (in Million IDR)	Rationale
Total Compliance Risk	~Rp400	Rp400	Potential fines and brand damage per untraced unit
Material Recovery Value	~Rp200	Rp200	Sale of sorted and high-quality SGTIN-verified recyclables
Reward Cost	(Rp150)	(Rp150)	Shopee Coins/vouchers issued upon verification
Logistics/Driver Bonus and QR	(Rp100)	(Rp100)	Incentive for Shopee Express collection and QR code printing
Platform/API Ops	(Rp50)	(Rp50)	Amortized cost of SGTIN database maintenance
Net Benefit / Unit	~Rp300	Rp300	Net positive gain (+)

5.2 Sensitivity Analysis

Scenario	Variable Change	Net Benefit (IDR)	Impact
Optimistic	53% collection traced and excellence quality waste	~Rp340	High volume reduces the per-unit amortized cost of API Ops.
Most Likely	44% collection traced and good quality waste	Rp300	Balanced rewards vs. material recovery.
Pessimistic	High reward to improve collection target	Rp200	If consumers require higher Shopee Coins to participate, profit margins tighten.
Risky	Low material value	Rp150	Quality of waste is low → the revenue drops.

Using 3-Points Estimation because it follows Beta distribution, we get

$$\mu = \frac{O + 4M + P}{6} = \frac{340 + 4 \times 300 + 200}{6} = 290 \text{ IDR}$$

5.3 Capital and Operational Expenditure

Financial Component	Estimated Cost (in Million IDR)	Rationale
Hardware Specification	Rp1.000	Upgrading production line printers to support variable SGTIN/QR for drivers data at high speeds
Initial API Integration	Rp300	Establishing the "Digital Bridge" between Nestlé SGTIN DB and Shopee's reward engine
Pilot Launch	Rp200	Initial marketing and training for Shopee Express drivers in the pilot region
Total Fixed Cost	Rp1.500	Total fixed cost and investment need for initial launching

5.4 Break-Even Point

The Break-Even Point is the volume of units (per SGTIN) that must be successfully traced and rewarded to recover the initial investment. We can calculate this by considering the mean value above in 5.2 Sensitivity Analysis and 5.3 Capital and Operational Expenditure (Capex and Opex).

$$\begin{aligned}
 Q_{\text{BEP}} &= \frac{\text{Fixed Cost}}{\text{Net per item}} \\
 &= \frac{\text{FC}}{\mu} \\
 &= \frac{1.500.000.000 \text{ IDR}}{290 \text{ IDR / Unit}} \\
 &\approx \mathbf{5.172.414 \text{ Units}}
 \end{aligned}$$

According to Purnamasari, et al., the most accessed and used feature in Shopee is free delivery, around 84.2% where Shopee has 53% Indonesian users. This means that at conversion rate, the percentage of Indonesians participating in this program is around 44% (from 84.2% of 53%). By using this number, Nestlé must produce 11.755.486 units (from $Q_{\text{BEP}} / 0.44$) or **12.000.000 units** to reach the break-even point. This is feasible within a year for a large manufacturing company like Nestlé.

Chapter VI: Environmental Impact and Scalability

6.1 Environmental Impact (Triple Bottom Line)

Category	Impact Metric	Explanation
Planet	Waste Reduction	Reduced untraced waste up to 53% at best-scenario and 44% at “conversion rate” by closing the “Traceability Gap”
	Circular Economy	Supports Nestlé's goal of Net Zero emissions by 2050 by ensuring plastic packaging is reintegrated into the production cycle rather than landfills
People	Consumer Habits	Leverages behavioral insights to transform recycling from a high-effort "chore" into a rewarding, seamless digital habit
	Community Engagement	Empowers the "Green Generation" to take active roles in sustainability through gamified engagement and tangible rewards
Profit	EPR Compliance	Directly addresses Extended Producer Responsibility (EPR), turning regulatory pressure into a competitive advantage for Nestlé
	Data Insights	Provides real-time, post-consumer data on recycling volumes and GTIN-level lifecycle tracking

6.2 Scalability

To implement the 4.2.3 Pilot-to-Scale Transition Strategy and ensure the solution is feasible and relevant for Indonesia's infrastructure, it is built on universal standards. By utilizing SGTIN (GS1 standards), the architecture is not proprietary to Nestlé nor a single manufacturing company, thus it can be adopted by any manufacturer in the Indonesian FMCG industry (GS1). In addition, leveraging Shopee Express, which handles a massive share of Indonesian e-commerce, provides a ready-made logistics backbone that can scale across various islands and cities. Once the pilot with Nestlé is proven, the "Eco-Return" checklist can be opened to other manufacturing partners, allowing them to "plug into" the existing Shopee-Waste Management network. This creates a national circular ecosystem.

Chapter VII: Conclusion and Key Answers

This "Trace & Reward" proposal presents a transformative solution to the growing environmental crisis caused by e-commerce packaging waste in Indonesia. By resolving the traceability gap that occurs once a product leaves the point of sale, this innovation provides Nestlé and other manufacturing companies with a viable path toward a truly circular economy.

Manufacturing companies can achieve not by redesigning their excellent package design, but by embedding waste management into existing consumer habits because the problem is not in the package design. "Trace & Reward" leverages the pre-existing Shopee app and Shopee Express logistics. For the consumer, "convenience" is maintained because the recycling process starts with a simple smartphone scan of a product they already bought. Affordability is protected as the system uses existing industrial standards (GS1), and "ease of use" is guaranteed by the intuitive digital interface, thus making the responsible disposal of waste as easy as ordering the product itself.

Furthermore, instead of traditional awareness campaigns, this approach uses Shopee's attraction to turn waste collection into a "game". In terms of delivery systems, the "Eco-Return" feature utilizes the reverse logistics of Shopee Express, allowing couriers to pick up waste during their regular routes. By designing packaging with unique SGTINs, we transform a static wrapper into a digital asset that holds value for the consumer once scanned.

The technology, SGTIN, provides the "ID" for every package, ensuring data-driven traceability. The collaboration between Nestlé (manufacturer), Shopee (platform), and Waste Management companies creates a closed-loop system that no single entity could achieve alone. Not only that, the system uses "instant gratification" (points/coins). By rewarding the user after the scan, we use "classical conditioning" to transform recycling from a chore into a rewarding habit.

Companies like Nestlé provide sustainable packaging and the "anchors" of the SGTIN data, and platforms like Shopee act as the digital bridge and logistics backbone, making the system accessible to millions. This leads to customers acting as the active "collectors" and "sorters" who drive the system through their engagement, where Waste Managers act as the final destination for recovered materials, ensuring they are processed back into the economy.

In conclusion, "Trace & Reward" is more than just a recycling program. It is a scalable ecosystem designed for the digital age and Indonesian's behavior. By aligning the incentives of manufacturers, e-commerce platforms, and consumers, this project paves the way for a future where every delivery brings us one step closer to a waste-free Indonesia, heavily supporting sustainability.

References

- Braham, Abraham Alex. "Indonesia's Internet Challenges." *Copyright Certificate*, October 2023, <https://copyright-certificate.byu.edu/news/indonesias-internet-challenges>. Accessed 27 December 2025.
- "Classical conditioning - Wikipedia." *Wikipedia, the free encyclopedia*, https://en.wikipedia.org/wiki/Classical_conditioning. Accessed 27 December 2025.
- Frigo, Giulia, et al. "Where does plastic waste go? Local dynamics of waste flows in Indonesian neighbourhoods." *Environmental Challenges*, vol. 19, 2025, <https://www.sciencedirect.com/science/article/pii/S266701002500054X?via%3Dihub>. Accessed 27 December 2025.
- IDN Times. "Indonesia Millennial and Gen Z Report 2025." *IDN Times*, <https://cdn.idntimes.com/content-documents/indonesia-millennial-genz-report-2025.pdf>. Accessed 25 December 2025.
- Indonesia. *Peraturan Menteri Lingkungan Hidup dan Kehutanan Republik Indonesia Nomor 75 Tahun 2019*. 2019. Accessed 24 12 2025.
- Ling, Jonathan. "What is the difference between GTIN, LGTIN and SGTIN?" *TrackVision AI*, <https://trackvision.ai/blog/what-is-the-difference-between-gtin-lgtin-and-sgtin>. Accessed 26 December 2025.
- "Mereka yang Tidak Memilah Sampah: 71,3% Merasa Merepotkan dan Bukan Tanggung Jawab." *GoodStats*, 30 November 2024, <https://goodstats.id/article/mereka-yang-tidak-memilah-sampah-67-3-merasa-merepotkan-dan-bukan-sebuah-tanggung-jawab-OgWBK>. Accessed 25 December 2025.
- Nestlé. "Plastic packaging and waste reduction." *Nestlé*, <https://www.nestle.com/sustainability/waste-reduction>. Accessed 23 December 2025.
- "Perverse incentive - Wikipedia." *Wikipedia, the free encyclopedia*, https://en.wikipedia.org/wiki/Perverse_incentive. Accessed 27 December 2025.
-



Purnamasari, Intan, et al. "Klasifikasi Pengguna Shopee Berdasarkan Promosi

Menggunakan Naïve Bayes."

<https://ojs.unpkediri.ac.id/index.php/gj/article/download/15998/2131/21521>. Accessed 26 12 2025.

"Shopee Jadi E-Commerce yang Paling Banyak Diakses di Indonesia 2025." *GoodStats*, 14 August 2025,

<https://goodstats.id/article/shopee-jadi-e-commerce-yang-paling-banyak-diakses-di-indonesia-2025-3COz0>. Accessed 25 December 2025.

"Standards." *GS1*, <https://www.gs1.org/standards>. Accessed 27 December 2025.

"What Are Common Barriers to Recycling? → Question." *Pollution → Sustainability Directory*, 30 November 2025,

<https://pollution.sustainability-directory.com/question/what-are-common-barriers-to-recycling/>. Accessed 27 December 2025.

Appendix

Appendix A: Return on Investment and Net Present Value

Here is the yearly prediction for the return on investment in this solution. At first, only a few people participated. However, as the year goes on, the number of people participating in this program will increase, gaining more traced and collected waste.

Scenario	Unit Volume Sold	Waste Traced and Collected	Net Revenue (in Million IDR)	ROI Annual Prediction
Break-Even Point	12.000.000	5.280.000	Rp1.500	0%
Mid-Scale Pilot	50.000.000	22.000.000	Rp6.380	~ 325%
National Expansion	100.000.000	44.000.000	Rp12.760	~ 750%

Here is the net present value calculated. Assuming the break-even point gets reached at the first year after the small-pilot.

Year	Phase	Waste Collected	Cash Flow (in Million IDR)	Net Present Value (in Million IDR)
0	Initial Investment	0	(Rp1.500)	(Rp1.500)
1	Break-Even Point	5.280.000	Rp1.500	Rp0
2	Growth	11.000.000	Rp3.190	Rp2,636
3	Mid-Scale Pilot	22.000.000	Rp6.380	Rp4,793
4	National Expansion	44.000.000	Rp12.760	Rp8,715
5	Sustained Growth	75.000.000	Rp21.750	Rp13,505
Total			Rp44.111	Rp29.650

Appendix B: SGTIN Implementation

The **SGTIN** is a specific type of Electronic Product Code (EPC) that adds a unique serial number to a standard **GTIN** (Global Trade Item Number). This transformation turns a generic product category into a "Serialized Waste Identity" (like most of the pharmacy products) that can be tracked through the Shopee ecosystem.

In digital systems, SGTIN data is represented in binary (0s and 1s) to ensure high-speed processing and compatibility with RFID or QR-based scanning. The most common industry standard is SGTIN-96, which uses 96 bits.

By this implementation, the global "maximum" product that can be identified is $2^{38} = 274.877.906.944$ products, which is a big number.

Appendix C: Behavioral Economics

To increase consumer engagement and conversion rate in Trace & Reward systems, manufacturing giants like Nestlé can leverage behavioral economics by aligning digital tracking (like SGTIN) with the deeply held values of religious or political cohorts. By utilizing nudges and framing, companies can transform a mundane task like scanning a QR code into a meaningful act of identity expression.

For religious demographics, the "traffic" in rewards programs can be boosted by framing waste management not as a technicality, but as a moral obligation.

The Concept: In many faiths, the concept of Stewardship (e.g., Khilafah or Amanah in Islam, or Creation Care in Christianity) dictates that humans are temporary caretakers of the Earth. Nestlé can use SGTIN data to provide "Environmental Tithes" or "Mizan Points." When a consumer scans a serialized product to recycle it, the app could display a message, "You have fulfilled your role as a steward by returning this resource to the circle." The reward is not just the digital token, but the psychological satisfaction of aligning with divine command.

For politically active segments, traceability can be framed as a matter of National Security or Economic Sovereignty. Political identity often thrives on in-group loyalty and the protection of local resources. In regions with high political polarization, the reward traffic can be incentivized by showing how serialized waste identities prevent local landfills from being filled by foreign untracked waste. Gamification can include regional leaderboards where different provinces or political districts compete to have the highest "Traceability Score." This taps into social norms and collective efficacy. Users scan and participate because they want their side or their community to be seen as the most disciplined and self-sufficient.